University Institute of Engineering

Department of Computer Science & Engineering

Experiment: 9

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Subject Name: DBMS

1. **Aim of the practical:** To create and implement PL/SQL packages by developing a package specification and package body containing procedures and shared cursors, in order to achieve modular, reusable, and efficient database programming.

2. Tool Used:

- **Database Management System:**
  - **PostgreSQL**
- **Database Administration Tool:**
  - **pgAdmin**

3. **Objective:**

To design and implement a PL/SQL package that includes procedures and shared cursors for structured and modular program development.

4. Practical / Experimental Steps

Step 1 :Create the base table (Employee) with appropriate attributes such as employee ID, name, department, and salary.

Step 2 : Insert sample data into the table to simulate real-world employee records.

Step 3: Design a package-like structure by creating a schema (in PostgreSQL) or package specification (in Oracle) to declare procedures/functions.

Step 4: Implement procedures/functions to fetch and display employee details.

Step 5: Define and use a shared cursor (or cursor logic) to retrieve employee records efficiently within the procedures.

Step 6: Execute the created procedures/functions to display all employee records and filtered results (e.g., by department).

5. I / O Analysis

```
CREATE TABLE employee (
   emp_id INTEGER PRIMARY KEY,
   emp_name VARCHAR(50),
   department VARCHAR(50),
   salary NUMERIC
);
```

```
CREATE TABLE

Query returned successfully in 126 msec.
```

```
INSERT INTO Employee VALUES (101, 'Amit', 'HR', 30000);
INSERT INTO Employee VALUES (102, 'Riya', 'IT', 50000);
INSERT INTO Employee VALUES (103, 'Karan', 'Finance', 45000);
INSERT INTO Employee VALUES (104, 'Neha', 'IT', 55000);
```

```
INSERT 0 1

Query returned successfully in 45 msec.
```

CREATE SCHEMA emp_package;

```
CREATE SCHEMA

Query returned successfully in 37 msec.
```

```
CREATE OR REPLACE FUNCTION emp_package.get_all_employees()
RETURNS VOID AS $$
DECLARE
   emp_rec RECORD;
BEGIN
   FOR emp_rec IN
      SELECT * FROM employee
   LOOP
      RAISE NOTICE '% % % %',
         emp_rec.emp_id,
         emp_rec.emp_name,
         emp_rec.department,
         emp_rec.salary;
   END LOOP;
END;
$$ LANGUAGE plpgsql;
```

```
CREATE FUNCTION

Query returned successfully in 109 msec.
```

```
CREATE OR REPLACE FUNCTION emp_package.get_emp_by_dept(p_dept VARCHAR)
RETURNS VOID AS $$
DECLARE
   emp_rec RECORD;
BEGIN
   FOR emp_rec IN
      SELECT * FROM employee
   LOOP
      IF emp_rec.department = p_dept THEN
         RAISE NOTICE '% % % %',
```

```
            emp_rec.emp_id,
            emp_rec.emp_name,
            emp_rec.department,
            emp_rec.salary;
        END IF;
    END LOOP;
END;
$$ LANGUAGE plpgsql;
```

```
CREATE FUNCTION

Query returned successfully in 38 msec.
```

OUTPUT

SELECT emp_package.get_all_employees();

```
NOTICE:  101 Amit HR 30000
NOTICE:  102 Riya IT 50000
NOTICE:  103 Karan Finance 45000
NOTICE:  104 Neha IT 55000

Successfully run. Total query runtime: 84 msec.
1 rows affected.
```

SELECT emp_package.get_emp_by_dept('IT');

```
NOTICE:  102 Riya IT 50000
NOTICE:  104 Neha IT 55000

Successfully run. Total query runtime: 62 msec.
1 rows affected.
```

6. Learning outcomes (What I have learnt):

- Understand the concept and working of materialized views in a database system.
- Differentiate between simple views, complex views, and materialized views.
- Create and refresh materialized views in PostgreSQL.
- Measure and compare query execution time for different types of views.

- Analyze performance benefits of materialized views in data-intensive applications.
- Apply materialized view concepts in real-world company scenarios such as SanDisk, JTG, and PayPal.